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Monitoring the COVID-19-affected Indonesian seaweed industry using remote sensing data

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ABSTRACT

The Indonesian seaweed industry has rapidly expanded over the last twenty years and now contributes to the livelihoods of 267,000 coastal households. Policy makers need accurate information from which to base policy that improves the livelihoods of smallholder farmers without further straining natural resource limits. The sustained growth of the sector necessitates means for reporting and monitoring the sector, particularly in the context of COVID-19. This paper uses newly available, high resolution, high frequency satellite data from April 2017 to December 2020 to identify seaweed farming plots and to map the seasonal changes in the industry. This is overlaid with data on environmental factors, prices, and official production statistics to reveal a range of insights into the industry which are obscured by official annual data, such as the seasonality of production and the change in activity since the beginning of the COVID-19 pandemic. The paper evaluates the usefulness of this data for monitoring the industry and informing strategic policy decisions.

1. Introduction

Over the last 20 years, seaweed farming has expanded rapidly across coastal Indonesia. Although many species of seaweed are commercially cultivated, just seven of these represent over 95% of the value of global aquatic plants [13]. Of these, four are high value food products grown primarily in China, Korea and Japan. The remaining three – species of *Gracilaria*, *Eucheuma* and *Kappaphycus* – are used to produce carrageenan and agar, which are hydrocolloids used as gelling agents in a range of applications. Indonesia produces 66% of global supply of these hydrocolloid seaweeds, the industry having expanded rapidly since the year 2000 (Fig. 1). Hydrocolloid seaweeds are much lower value than food seaweeds.

Indonesian seaweed production is undertaken mainly by low income smallholder farmers, primarily in Eastern Indonesia, who are able to work around seasonal and tidal changes and have limited other income generation opportunities. Seaweed farming is an important source of income for over 267,000 rural households in Indonesia (Presidential Decree 33-2019, p.17), who earn profits of around US$2,000 annually (Presidential Decree 33-2019, p.16). Farmers frequently perceive that seaweed farming is a more lucrative activity than more established industries such as copra, cocoa and other marine industries [3,5,23,35]. In addition, a range of other benefits from seaweed farming have been reported, such as an increased ability to manage their income and save for large purchases [23,35]. These benefits sit alongside a variety of other environmental and social changes associated with the industry [4,28,34,38,40] and varying benefits to different community and household members (see, for example, Refs. [11,17,21]).

In recognition of the increasingly important role of seaweed in coastal livelihoods, the Indonesian government has made the industry development a national priority, most recently seen in Presidential Decree 33-2019 (*Peraturan Presiden Republik Indonesia Nomor 33 Tahun 2019 tentang Peta Panduan Pengembangan Industri Rumput Laut Nasional Tahun 2018–2021*), which provides a road map for the development of the seaweed industry from 2018 to 2021. The decree outlines opportunities for increased seaweed production and, rather than exporting early-stage products, developing processing capacity within Indonesia of food products, animal feed products, fertilisers, cosmetics and bioethanol (e.g. [1,22,27,30,33,36,37,39,41]). Efforts to improve production focus on growth rates and quality improvements but note that a range of issues face the industry, such as high transportation and logistics costs and a lack of infrastructure development (Presidential

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Decree 33-2019 p. 30). Despite its importance to the national economy and as a poverty reduction strategy, the patterns of seaweed production are poorly understood, limiting the ability of farmers, traders and processors to coordinate and the efficacy of policy interventions [42,43]. There is therefore a need for more informed data to understand patterns of farming in rural Indonesia.

South Sulawesi (*Sulawesi Selatan*/Sulsel) is the largest seaweed producing province in Indonesia, producing over a third of the nation's seaweed, and more than 20% of the global supply of carrageenan seaweeds (Fig. 2). Biophysical characteristics of seaweed farming vary along the South Sulawesi coastline, where there are small bays, beaches, offshore islands, and estuaries, all of which affect seaweed farming geographies. In this paper, we use newly available, high resolution satellite data to map the production of seaweed. Data from this analysis highlights the extreme seasonality of production, and the importance of biophysical features in shaping village social and economic engagement with seaweed. While official production estimates are available on a quarterly basis, the use of this newly available, high resolution, high frequency data contributes a new source of information on seaweed farming patterns which can be triangulated with existing production estimates to inform policy decisions. Since seaweed farming cycles are typically 40 days, quarterly estimates include two cycles, obscuring monthly changes in production. This satellite data can be usefully employed to inform policy management of coastline resources and to remotely detect changes in seaweed production, such as those associated with farmer extension programmes, price fluctuations or global economic events such as the COVID-19 pandemic.

2. Materials and methods

This study applies newly available satellite imagery to monitor seaweed production in Indonesia. It does so using a case study of the mainland coastline of the *Pangkajene dan Kepulauan (Pangkep) Regency*[1] (Fig. 3). This location is an established seaweed production region exhibiting high seasonality of production. The case study is used to develop a method for the use of satellite data to monitor seaweed production patterns, to assess the usefulness of the data generated. This study tests the feasibility of the method, which can later be applied to larger areas such an entire province, to support improved monitoring and coordination of the industry in those areas. This data was triangulated with rainfall data obtained from the *Badan Meteorologi, Klimatologi, dan Geofisika* (BMKG) [6] online data service (collected from the *Stasiun Klimatologi Maros*, the weather station nearest to Pangkep). Daily rainfall data in millimetres was obtained and aggregated to obtain monthly values.[2] Price data was collected from Jasuda (Jaringan Sumber Daya) from 2017 to 2020 [19]. Quarterly production estimates were provided by the South Sulawesi Department of Marine Affairs and Fisheries (*Dinas Kelautan dan Perikanan*/DKP). Informal interviews were undertaken over the phone with existing contacts in the seaweed industry, including some farmers (who provided price data and other observations) and industry professionals. This information was used to inform interpretation of the data.

The use of satellite imagery to monitor seaweed production is currently limited. Seaweed culture sites are not visible on low-resolution Landsat satellite imagery [18]. However, they are clearly visible on newly available, high-resolution satellite imagery such as that available from PlanetLabs, which offers a resolution of 1 pixel to 4.77 m (Fig. 4). This high-resolution, high-frequency satellite data has only recently become available. PlanetLabs have produced such imagery since 2016 [31], and have satellites collecting new images generally every few days. In 2019, they began producing weekly, monthly and quarterly basemaps from this imagery. Basemaps are mosaics of images taken across the month and pieced together to account for cloud cover. The high frequency of collection means that despite frequently high cloud cover in the rainy months, a complete mosaic of the study area is available each month. These basemaps are available from 2016, although basemaps for data prior to April 2017 were not of sufficient quality for this analysis due to high cloud cover likely resulting from insufficient frequency of imaging. In this study, monthly basemaps were used to identify plots of farmed seaweed along the mainland coastline of Pangkep. Shapefiles were created using rectangles to represent each plot of seaweed, and maps were generated to show the area under seaweed production in each month. The area of these rectangles was aggregated to calculate the total area under seaweed production in each month.

The satellite data provided here has limitations. The accuracy of the assessment is limited by the resolution of the imagery, and the presence of cloud cover which reduces the visibility of the ocean surface. This study used monthly basemaps, which are mosaics of images taken over the month; as such they represent a range rather than a point of time. Further, while it is possible to identify seaweed plots with reasonable accuracy in shallow water, this becomes more difficult in deeper water. Coastal areas differ from landcover assessments because there is a colour gradient in the coastal waters, which introduces challenges for automated image processing and makes processing of imagery over large areas difficult, since plots are identified by their shape more than by their colour. Development of a method for automatic processing of this imagery would vastly increase the feasibility of using this data source over a large area.

3. Results

3.1. Spatially and temporally detailed production estimates

Fig. 5 shows the results of the mapping exercise for the year of 2018. 2018 was selected as a typical year as 2019 received atypical rainfall resulting in lower production while in 2020 the COVID-19 pandemic has affected production dynamics. The image shows the extreme seasonality of seaweed farming, with over two thirds of production occurring in just three months of that year (Feb-April), and 90% of production occurring in the first 6 months of that year (Fig. 5).

The seaweed production data derived from satellite imagery is overlaid with rainfall patterns in Fig. 6. Seaweed growth rates are strongly affected by seasonal variations in salinity, light intensity and sea waves. In particular, the salinity of the water has a major effect on seaweed cultivation. Neish [29] reports that growth rates at favourable times of year are as much as five times higher than in less favourable times. The hydrocolloid seaweeds grown in Indonesia grow best at salinities of 29–34 ppt (Presidential Decree 33-2019, p. 22). This is slightly lower salinity than the average seawater salinity, which is 35 ppt. Seaweed growth rates are strongly affected by salinity levels in the water, which are closely linked to regional rainfall patterns [7]. From March to May, salinity in the Java sea offshore of South Sulawesi is substantially lower than the rest of the year as a result of high rainfall in

[1] Pangkep also contains 115 offshore islands which may contribute to seaweed production in the regency, however quantities are likely to be low due to the higher salinity and greater remoteness of these islands. Satellite imagery of all islands was reviewed and seaweed production was identified offshore of four islands off the Pangkep mainland and has been included in the results (*Pulau Banabanawang*, *Pulau Gusungtinggi*, *Pulau Sabangko* and *Pulau Salemo*). In addition, small amounts of seaweed offshore of *Pulau Sabalana* was identified but could not be included in the analysis due to difficulty accurately measuring seaweed plots in that marine environment. In addition, some shapes were observed offshore of *Pulau Marasende*, but these were small in size and it was unclear whether these were small amounts of seaweed (potentially growing on bamboo rafts) or other structures (such as fish traps) and so they could not be included in the analysis. It is possible that seaweed production on other islands occurs but could not be identified using the satellite imagery. This highlights the importance of ground truthing work to ensure that remote seaweed producers are not disadvantages by this approach.

[2] Unmeasured days were common but no adjustments were made to account for this.

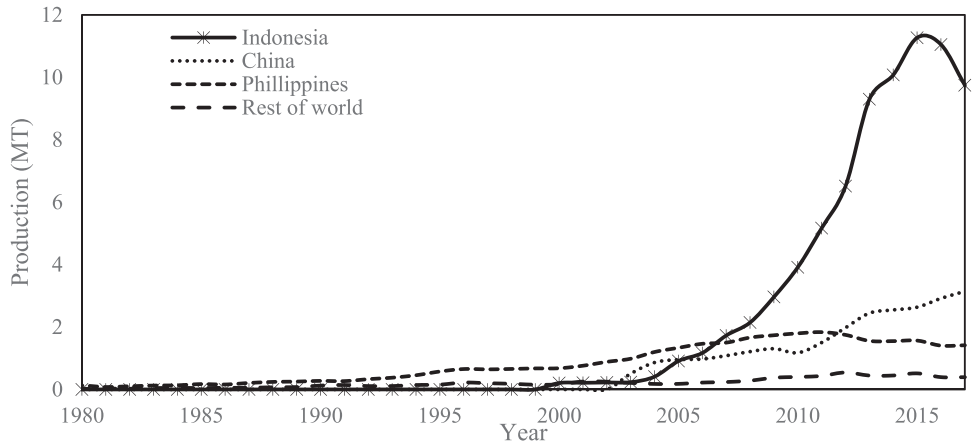

Fig. 1. Global production of major hydrocolloid seaweeds.
Source: FAO [13].

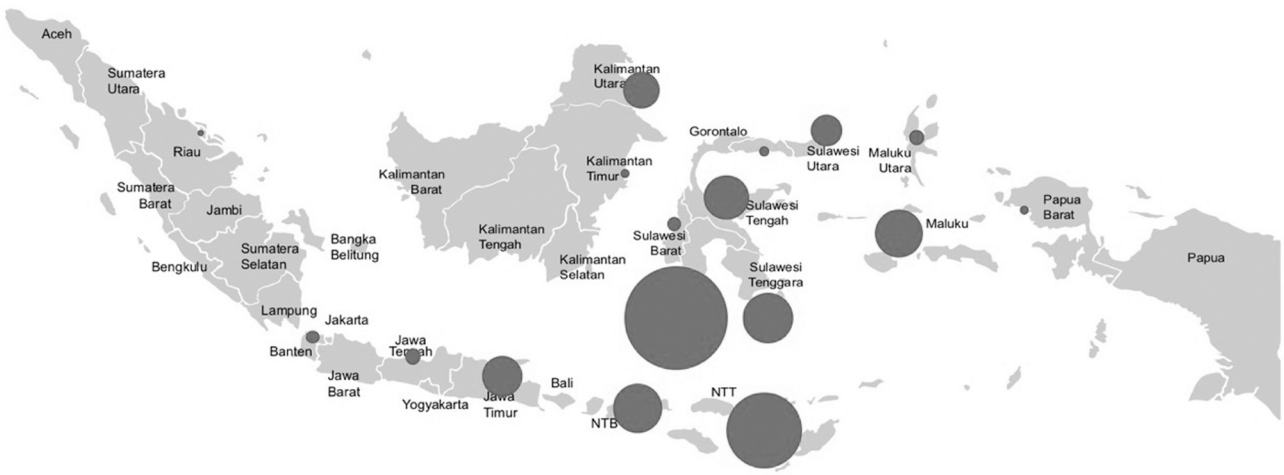

Fig. 2. 2017 Seaweed production in Indonesian provinces by weight.
Data source: Ministry of Marine Affairs and Fisheries, Republic of Indonesia ([26]: 320–321).

surrounding areas [7]. Rainfall in South Sulawesi also increases river flows that transports nutrients to the sea [15], although excessive rainfall may inhibit seaweed growth. During periods of favourable freshwater flows in the first half of the year, seaweed farming expands to cover offshore areas, and contracts again as river flows reduce and seawater enters the Java sea later in the year.

Access to these detailed maps of production throughout the year offers substantial opportunities to industry and government to improve their co-ordination with farmers by better understanding their activities and needs. Remote sensing can provide important data for improved monitoring of coastal areas and more sustainable use of resources [25] and several countries are using satellite imagery to monitor coastal ecosystems and aquaculture activities [24]. Currently the primary use of satellite imagery for aquaculture is in site selection, however there is scope for broader use in monitoring of production patterns. Data could be used to monitor production patterns to identify the needs of farmers throughout the year, including in transport and storage, inputs (such as seeds), and extension services. Downstream actors (traders and processors) would be interested in information on supply volumes and quality attributes to plan for aggregation, transport and storage. The data could also inform marine planning and zoning, including applications in monitoring expansion of production and production in areas which are protected or have multiple uses (such as for fishing and sea access).

3.2. Triangulation with official production estimates

The seasonal variation in seaweed farming in Pangkep is well known to farmers, industry and government, however, government faces considerable difficulty in generating accurate estimates of production. It may be possible to use satellite imagery to generate production estimates by converting area observations to approximate production weight estimates. This may provide cross-verification with current statistical methods estimated by the DKP. DKP quarterly estimates roughly correspond to two growing cycles, and are generated by surveyors who work in villages and interview farmer groups, who report the number of ropes currently in production. The reporting period for the quarter is in the two weeks following the end of the quarter, and a sample of farmers is taken (rather than interviewing all farmers). The number of ropes reported is converted to an area estimate by assuming a 1 m spacing between ropes and multiplying this by the rope length. This is converted to a weight estimate by assuming that farmers use about 280 g of propagule per metre of rope, and that this results in a harvest of 2.24 kg wet seaweed per metre of rope. This translates to a yield of 22.4 tonnes of wet seaweed harvested per ha under production. This figure is approximate and varies substantially with farming system, including key factors such as seed spacing, seaweed growth rates and initial propagule weight. Neish [29] also produces an area to harvest weight estimate based on farming systems in Madura. He noted that farmers used cuttings of 80–100 g spaced at 0.2 m intervals for a resulting

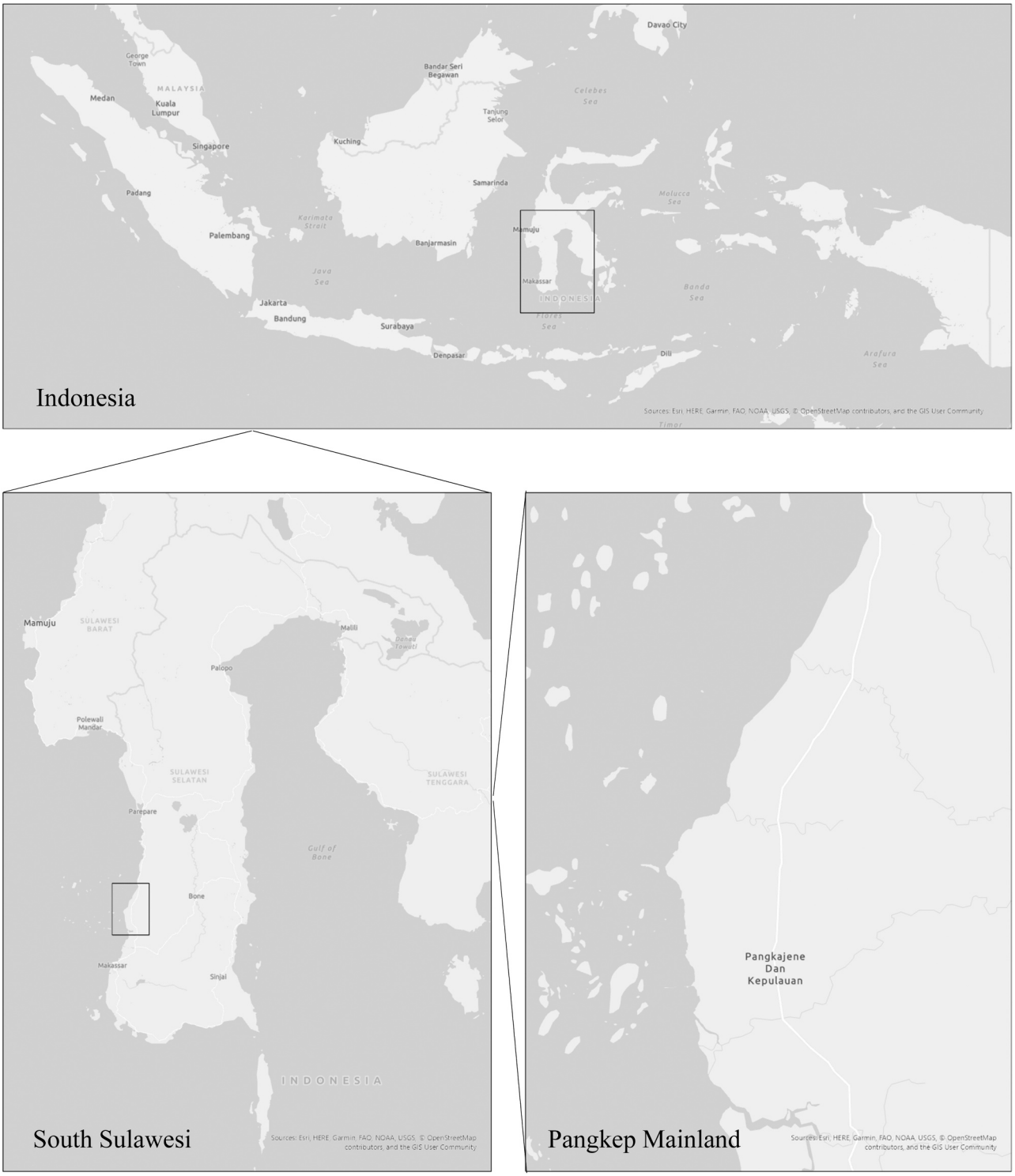

Fig. 3. Study site location.

weight of 0.4–0.5 kg of propagules per metre with a resulting harvest of two to four times the initial propagule weight, depending on the season. Taking the mid-point of 0.45 kg propagule per metre, a low season yield would produce 0.9 kg per metre and a high season of 1.8 kg per metre. Applying the assumption of 1 m spacings between ropes using in the Pangkep conversion generates an estimate of 9–18 tonnes of wet seaweed per ha, which is a lower range than that used for the Pangkep conversion but is comparable considering the diversity of farming systems across Indonesia. Converting our area/month observations to weight estimates using the 22.4 tonnes/ha figure employed by Pangkep officials, and converting months to cycles assuming a 30-day month and a 40-day cycle (0.66 cycles per month) allows for comparison with official data, as shown in Fig. 7.

Significant differences between the official production estimates and the satellite data are evident. Firstly, official estimates over the time-frame available are nearly ten times higher than what might be expected based on satellite imagery. There is a need for a survey of farm systems to test the validity of the assumptions on which the conversion is based,

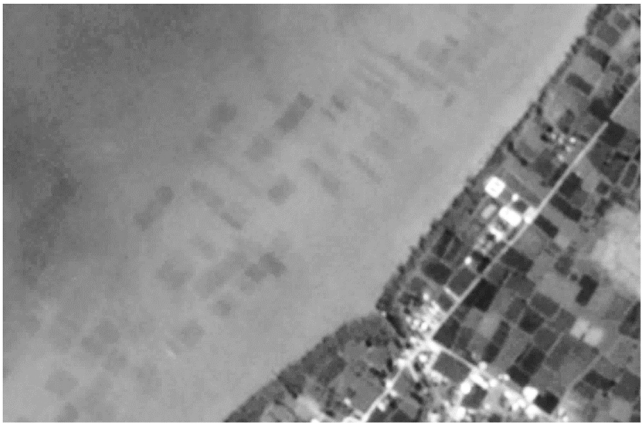

Fig. 4. Seaweed plots visible on PlanetLabs basemap at a resolution of 1 pixel to 4.77 m for May 2020 at 119.5683687°E 4.5996097°S.

and a review of data collection methods. Secondly, official production statistics show much less variation between quarters than that measured by satellite data and obscure seasonal variation. This suggests issues with the reporting of production by farmers, as production reports remain high throughout the year despite much smaller areas of seaweed under cultivation. It could also be related to sampling issues. In addition, official production estimates appear to have an approximately three-month time lag from that estimated by satellite data. This means that data reported for each quarter is reported for being for three months *earlier* than suggested by satellite data. This suggests that there may be misreporting of time periods, for example if farmers asked to report a previous quarter's completed production actually report an estimate based on their current production.

These discrepancies suggests that satellite data could potentially be used to improve production estimates for a closer understanding of the seaweed production cycles in different areas of the province, particularly given that production patterns throughout the year vary with location [12]. Satellite imagery could be an important central government tool in providing an additional source of data for monitoring regional seaweed production, particularly in circumstances where resources for data collection are scarce or incentives for overestimating production exist.

However, two caveats on the potential of using satellite data to improve the accuracy of monitoring and reporting apply. First, thorough 'ground truthing' would be required to explore farming systems around Eastern Indonesia and growth rates in different locations in order to develop an accurate system for converting area observations to production volume estimates across diverse landscapes and seasons. This is necessary to refine the assumptions used and calibrate and ground truth the models of seaweed production. In addition, the mapping would need to be expanded across a large area, a potentially resource intensive task due to the difficulties processing imagery of coastal areas. Second, one species of seaweed, *Gracilaria*, can be grown in both marine water or inland ponds, and therefore would not be visible on satellite imagery of the coastline. Significant volumes of *gracilaria* are grown in Sulsel – in 2019, this species represented 28.4% of all seaweed production [12]. However, in Pangkep, this species is less widely cultivated, representing only 2.7% of production [12], and as such this is not likely to be a significant explanatory factor for the discrepancies observed. Third, the mapping has been undertaken only along the mainland of Pangkep, and therefore excludes any farming undertaken on Pangkep's numerous outer islands. The contribution from these islands is expected to be small however due to their remoteness and small size.

3.3. Monitoring the seaweed industry during COVID-19

A third useful application of this data is in providing insights into the seaweed industry during the COVID-19 global pandemic. The outbreak of the global pandemic for COVID-19 has had far reaching impacts globally. This has been accompanied by travel restrictions which limit the ability of researchers and government to explore the impacts of the pandemic. Satellite data offers insights into seaweed farming activities occurring during COVID-19. In 2020, the area under seaweed production from May to August as observed in satellite data was on average more than double that in previous years (Fig. 8). No increase was observed from Feb-April, likely because the area suitable for seaweed farming was already close to capacity in these months. The higher area under seaweed cultivation suggests either an increase in seaweed planting, or a decrease in seaweed harvesting. It is likely that an increase in seaweed planting is occurring as a result of a range of environmental, socio-economic and policy factors. Environmentally, 2020 is a La Nina year, and as a result, variations in rainfall and water temperature are expected. In addition, the COVID-19 global pandemic has had widespread impacts on employment across Indonesia, which could see increased labour availability to seaweed farming if people (particularly youth) who were previously working or studying in urban areas have turned to seaweed farming. In addition, agricultural extension policy may create change in the seaweed industry through changed farming techniques which make seaweed farming more favourable, such as government seed distribution programmes which were implemented in Pangkep in 2020. Alternatively, it is possible that a decrease in seaweed harvesting has occurred in response to low prices, either as a storage technique or due to insufficient returns to labour. Although field surveys are needed to interpret this data, reports suggest that an increase in production is occurring during the COVID-19 Public Health Emergency of International Concern (PHEIC) despite reduced government production targets [2].

A clear effect of the COVID-19 PHEIC has been a decline in seaweed prices alongside a decrease in the opportunity cost of labour due to reduced employment opportunities. In Bali, reports suggest that the collapse of the tourism industry due to the COVID-19 pandemic has led many tourism operators to return to seaweed farming, which many had left seeking better returns [9,32]. This is despite significant declines in seaweed farmgate prices (Fig. 9). On average, prices declined by 27% from January to December 2020 in South Sulawesi locations for which data is available. This trend is reflected throughout the country, with average prices in the six regions shown below dropping from 19,800 IDR to 14,700 IDR from January to December 2020 (Fig. 9). As seaweed farming in many areas was already a marginally productive activity, such a decline could completely erase the profitability of the activity after accounting for input, capital depreciation, and labour costs. A report from Sulawesi suggested that in one region, prices fell from 22,000 to 13,000 while seed costs remained constant, such that farmers reported suffering losses from seaweed farming after accounting for seaweed seed and labour costs [14]. An increase in production alongside a decrease in price, as observed here, is the inverse of the normal response. One explanation for this is lower input costs, in this case likely that of labour as a result of lower opportunity costs resulted from reduced income generation opportunities off-farm. The decrease in seaweed prices is suggested to be linked to temporary export suspensions during the COVID-19 PHEIC, leading to a lack of storage facilities and falling prices [20]. These factors suggest significant disruption to the seaweed industry as a result of the COVID-19 PHEIC.

Satellite imagery analysis could potentially be employed in the industry to monitor significant changes in production, which, alongside other data such as price data, could provide insights into effects of global events on the livelihoods of farmers. The COVID-19 PHEIC is one example of a situation in which monitoring of the industry remotely could be a beneficial tool.

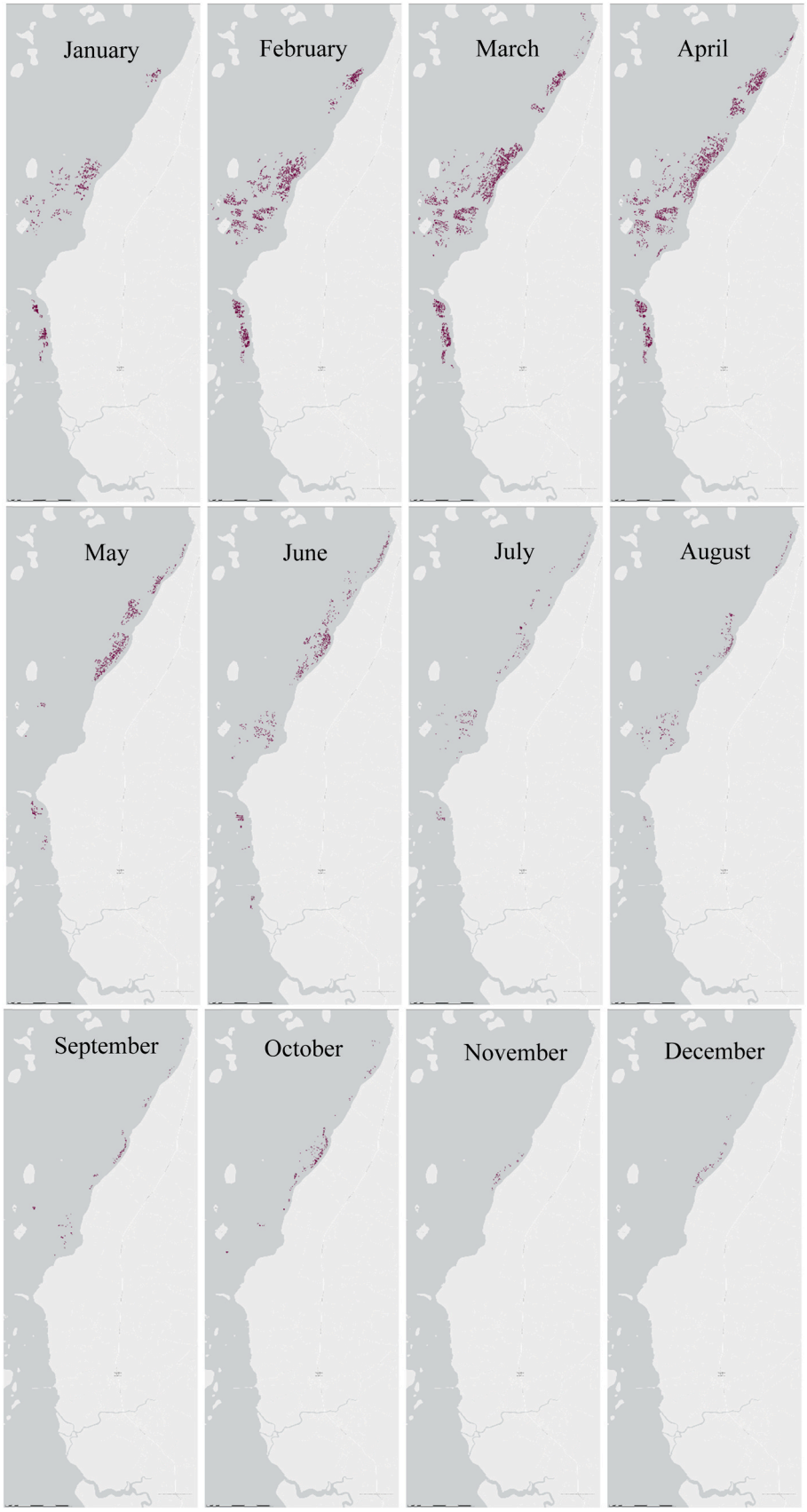

Fig. 5. The area of seaweed planted along the mainland coastline of Pangkep, Sulsel in 2018.

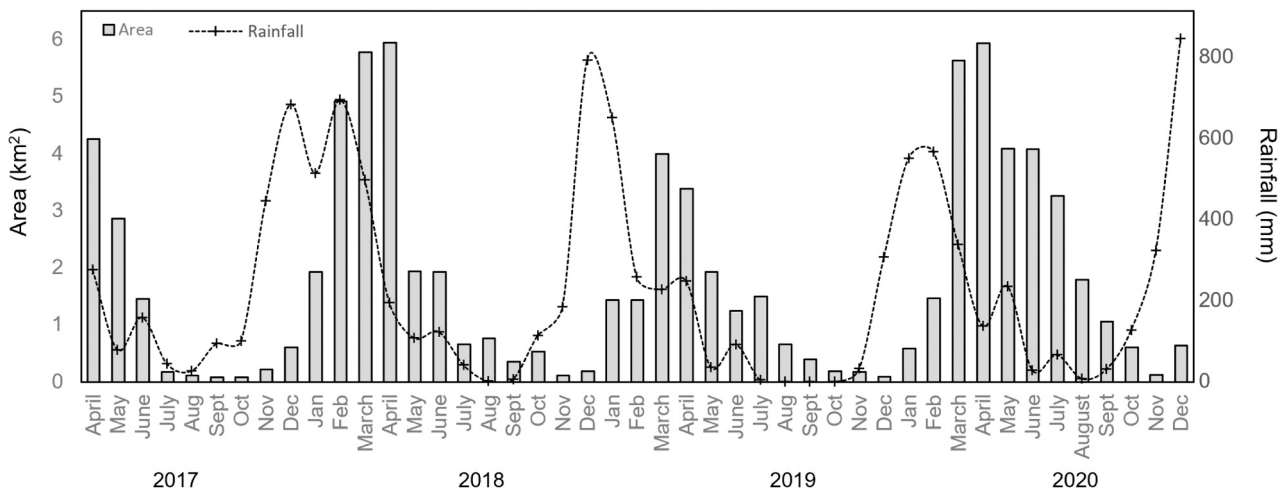

Fig. 6. Area under seaweed production as viewed on satellite data mapped against rainfall. Production is highly seasonal and linked to rainfall patterns. Data Sources: BMKG [10]; Planet Team [31].

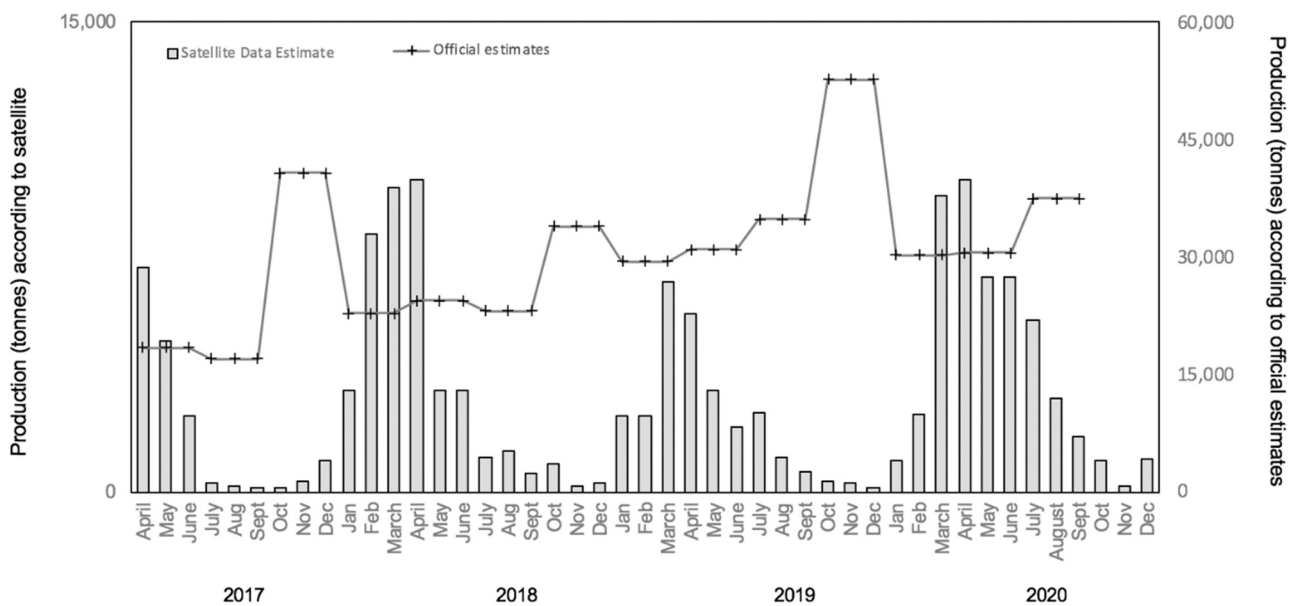

Fig. 7. Official production estimates are nearly ten times higher than production rates indicated by satellite data analysis. Data Sources: DKP Sulsel [12]; Planet Team [31].

4. Discussion

4.1. Applications of satellite imagery as a monitoring tool

The results of this analysis suggest that there is significant potential for newly available, high-resolution, high-frequency satellite data to be usefully employed as a reporting and monitoring tool for the Indonesian seaweed industry. A more accurate, and spatially and temporally detailed understanding of seaweed farming patterns is important for a range of purposes, such as monitoring production rates and patterns, land and sea use planning and zoning, improving national and international statistical reporting processes, responding to significant economic and environmental events, and for research and development to improve industry integration and efficiency.

Firstly, high resolution satellite data makes possible remote monitoring of coastal ecosystems and production systems in similar ways that lower resolution satellite imagery has been used to monitor changes in forestry [8]. This allows government at multiple levels to observe changes in seaweed production across Indonesian coastlines, including expansion into new areas and changes in production in existing areas. This data on production patterns could be combined with use of remote sensing data to monitor ecosystems to monitor any correlation between production changes and changes in ocean ecosystems.

Secondly, this data can be used for marine area planning. This may include assessing viable production sites for seaweed as has been done previously [24] and identifying sensitive marine areas for protection, shipping zones and fishing zones, and monitoring any encroachment on those areas.

Thirdly, this data can be used for cross verification of statistical reporting on seaweed production rates for better coordination between central and regional governments. There are a range of ways that inaccuracies are introduced into regional statistics, including insufficient resourcing. Satellite data, if combined with location-specific information on farming systems, environmental features and events, and phyconomic characteristics, could be used for reporting of production rates at a greater accuracy, which would support more effective policy

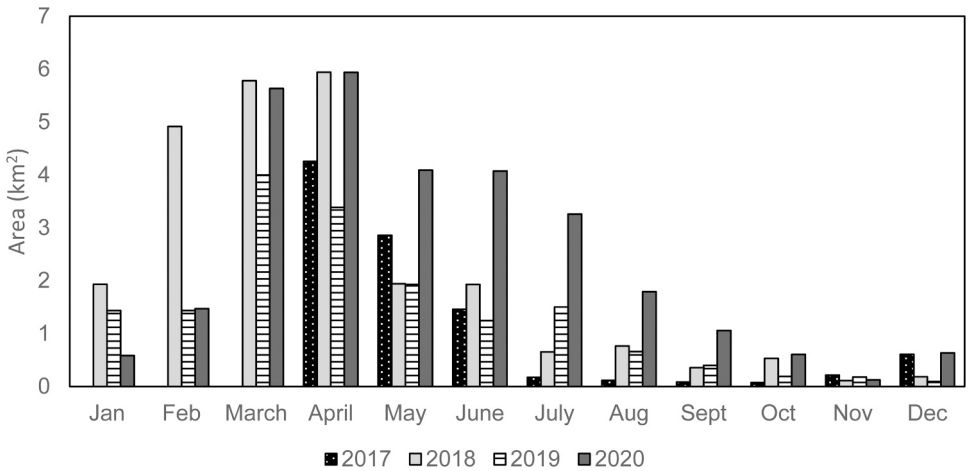

Fig. 8. Area under seaweed production from May – August 2020 is significantly higher than in previous years (satellite data).

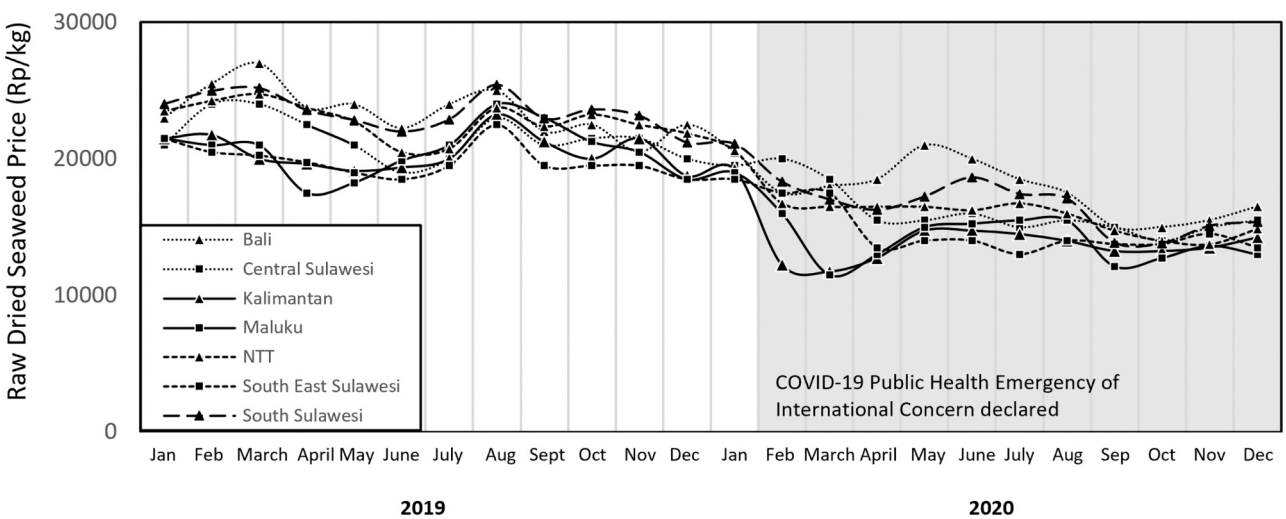

Fig. 9. Purchase price for dried cottoni (Kappaphycus Alvarezii) across Indonesia have declined an average of 27% from January to December 2020. Data Source: Average monthly price reported on JasSuDa [19].

responses. This could potentially be done at a lower cost than existing surveys, depending on costs of data, data analysis, ground truthing work, and existing survey work.

Fourthly, this data can provide very fast and up-to-date information on the impact of significant environmental and socio-economic events, such as the COVID-19 PHEIC. During the COVID-19 PHEIC, researchers and many members of industry and government have been unable to travel to remote areas to assess the impact of the pandemic on seaweed farming communities, who are typically very low income and vulnerable to price fluctuations of key commodities. The ability to monitor production remotely offers one tool for evaluating the impact of the pandemic on farmers in different regions. This can help to identify areas where significant changes have occurred and to direct resources to areas in particular need.

Finally, this data provides very useful information to support research and development of the industry and to improve coordination and efficiency. Research in phyconomy would benefit from data on seaweed area of cultivation, production and seasonality to test effects against a range of environmental variables including water salinity, temperature, irradiance, wave action, fertiliser run-off, ocean currents, coastline geographies and marine life. Some studies have attempted to assess the relative importance of these factors [16], however, constraints on the size of the study and the number of locations considered limits the potential to assess a range of factors in a range of geographies. Understanding seasonal patterns of seaweed farming across a large area – for example, if satellite data were analysed for the whole of South Sulawesi, or at a number of locations throughout Indonesia –could be used to generate robust models of the likely impact of a range of environmental factors on seaweed farming.

In addition, the data could be used to support improved coordination in the industry, for example by providing industry (processors, traders, exporters) with detailed current estimates of seaweed in the water (prior to harvest) such that processing of the seaweed can be planned and efficiency increased. Improvements in processing are important for adding value to the industry for both improving processing profitability and increasing returns on farming to seaweed farmers, who are often low-income populations. Government could also support the improved coordination of the industry by using satellite data as inputs for logistics modelling across the province, which would vastly increase the accuracy of logistics models through the use of data which is detailed and disaggregated over short time spans and over very small spaces. This information would prove useful in planning transport networks to meet the needs of the seaweed industry.

4.2. Limitations

Any technique for measuring seaweed production has limitations, such that combining multiple sources of information can lead to a more accurate understanding of production. Official quarterly production estimates are based on farmer reporting of the number of ropes currently in use on a quarterly basis. This approach is useful in farmer estimates of their production. However, it has several limitations, such as aggregation of multiple seaweed farming cycles into a single estimate (obscuring production differences between them), the possible introduction of a time-lag, and a reliance on farmer estimates. The satellite data provided here similarly has limitations, notably that like official estimates, it also measures the variable of area under production rather than weight produced. Since the growth rate often varies by up to five times between seasons [29], seasonal production differences may in fact be more or less pronounced than estimated by this data, depending on whether farmers respond to lower growth rates by reducing production (and perhaps diverting their labour to more productive activities) or increasing production (as may occur if other income generation activities are not available). In addition, both the satellite data and the quarterly production estimates generate estimates of production based on area under cultivation, and as such fail to account for losses prior to harvest or during drying or storage. Additional research into storage practices of farmers, and triangulation with reports on buying quantities could improve understandings of seaweed value chains at the farmer level. Despite these limitations, the data presented in this paper provides a useful additional source of data for mapping seaweed cultivation which is easy to generate across large areas to a high level of detail. It can be applied in reproducible ways across large areas, and so could be usefully employed to benchmark other government production estimates and improve their accuracy, and as a point of comparison for a range of farming locations. This data therefore provides a very useful source of information for refining official estimates, as a cross-verification of sources and development of a database of production features in different locations is expected to lead to a more accurate picture.

5. Conclusion

Policy makers face challenges monitoring the seaweed industry and developing strategies for improved coordination between farmers and processors. This paper has demonstrated the utility of newly available, high resolution, high frequency satellite data in mapping the industry throughout the year. It takes initial steps to organise and interpret this new data set, with the view that expanding this analysis geographically could provide important data to facilitate better coordination between farmers and processors, planning of transport and logistics, research and development in seaweed production, and monitoring of the impact of important global events such as the COVID-19 PHEIC. This data could be usefully integrated into marine planning programmes for better characterisation of the industry to inform the development of strategic policy initiatives. Satellite imagery is indicated as a potentially highly useful tool for improved monitoring of the Indonesian seaweed industry. Further research developing this method could seek to improve image processing methods, expand mapping over larger areas and undertake fieldwork to collect ground truthing data.

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